

Question 1 13 Marks**Marks**

- (a) Find the primitive of $(3x-2)^{10}$ 2
- (b) The gradient of a curve is given by $\frac{dy}{dx} = 2x-3$. What is the equation of the curve if it passes through $(1,5)$ 2
- (c) The area enclosed by the curve $y = 3x - x^2$, the x -axis, the lines $x = 1$ and $x = 2$, is rotated about the x -axis. Find the volume of the solid formed. Give your answer in terms of π . 4
- (d) In what ratio does the x axis divide the area of the region bounded by the parabolas $y = 3x - x^2$ and $y = x^2 - x$? 5

Question 2 14 Marks *Start a new page*

- (a) Solve $\log_e(4-3x) = 2\log_e x$. 4
- (b) Differentiate
- (i) $\log_e(3-2x^2)$ 1
- (ii) $2x \ln 3x$ 2
- (c) If $y = x \log_e x$, find:
- (i) $\frac{dy}{dx}$. 1
- (ii) Find the coordinates of the stationary point. 2
- (iii) Determine the nature of the stationary point. 1
- (iv) Find the value of y (correct to three decimal places) when $x = 0.001$. 1
- (v) Use the information found to sketch the curve $y = x \log x$ 2

Question 3 13 Marks Start a new page Marks

(a) Find the value of $\int_{-2}^2 e^{2x} dx$ 2

(b) Differentiate e^{-x^2} . Hence show that 4

$$\int_0^1 x e^{-x^2} dx = \frac{1}{2} \left(1 - \frac{1}{e} \right)$$

(c) The number of bacteria, N , in a colony after t minutes grows according to the law $\frac{dN}{dt} = kN$, where k is constant.

(i) Show that $N = N_0 e^{kt}$, where N_0 is constant, is a solution of $\frac{dN}{dt} = kN$. 1

(ii) If the number of bacteria is doubled in 200 minutes, find the value of k . 1

(iii) How long, to the nearest minute, does it take for the number of bacteria to grow to 10 times the original number? 2

(d) Evaluate $\int_{-1}^1 x^2 e^x dx$ by using Simpson's rule with three function values. Answer correct to 2 significant figures. 3

End of Examination

Q1

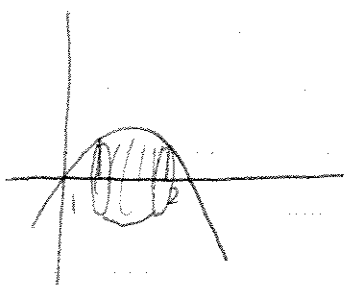
$$(a) \frac{1}{33} (300-2)^4 + C$$

$$(b) y = x^2 - 3x + C$$

$$x=1, y=5 \therefore C=7$$

$$y = x^2 - 3x + 7$$

(c)



$$V = \pi \int_1^2 (3x - x^2)^2 dx$$

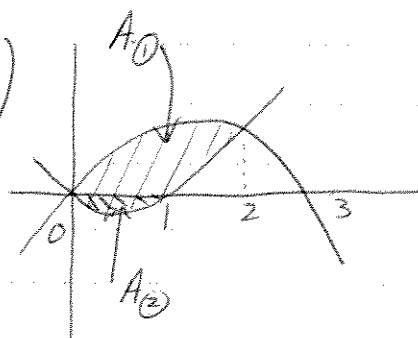
$$= \pi \int_1^2 (9x^2 - 6x^3 + x^4) dx$$

$$= \left[3x^3 - \frac{3}{2}x^4 + \frac{1}{5}x^5 \right]_1^2$$

$$= 3 \cdot 2^3 - \frac{3}{2} \cdot 2^4 + \frac{1}{5} \cdot 2^5 - \left(3 - \frac{3}{2} + \frac{1}{5} \right)$$

$$= 40.7 \text{ units}^3$$

(d)



$$3x - x^2 = x^2 - x$$

$$2x(x-2) = 0$$

$$x = 0, 2$$

$$A_1 = \int_0^2 (3x - x^2) dx - \int_1^2 (x^2 - x) dx$$

$$= \left[\frac{3}{2}x^2 - \frac{x^3}{3} \right]_0^2 - \left[\frac{1}{3}x^3 - \frac{x^2}{2} \right]_1^2$$

Q1 c+d

$$A_{①} = \left[\frac{3}{2} \cdot 2^2 - \frac{2^3}{3} - 0 \right] - \left[\frac{1}{3} \cdot 2^3 - \frac{2^2}{2} - \left(\frac{1}{3} - \frac{1}{2} \right) \right]$$

$$= 2\frac{1}{2}$$

$$A_{②} = \left| \int_0^1 (x^2 - x) dx \right|$$

$$= \int_1^0 (x^2 - x) dx$$

$$= \left[\frac{1}{3} x^3 - \frac{x^2}{2} \right]_1^0$$

$$= 0 - \left(\frac{1}{3} - \frac{1}{2} \right)$$

$$= \frac{1}{6}$$

$$A_{①} : A_{②} = 2\frac{1}{2} : \frac{1}{6}$$

$$= 15 : 1$$

Q2

(a) $\log_e(4-3x) = \log_e x^2$, $x > 0$ from original question

$$4 - 3x = x^2$$

$$x^2 + 3x - 4 = 0$$

$$(x+3)(x-1) = 0$$

$$x = -3 \text{ or } 1, \quad x > 0$$

$\therefore x = 1$ is only solution

(b) (i) $\frac{-4x}{3-2x^2}$

(ii) $2x \cdot \frac{x}{3x} + 2 \ln 3x$
 $= 2(1 + \ln 3x)$

(c) $y = x \ln x$

(i) $y' = \ln x + 1$

(ii) $y' = 0$ when $x = \frac{1}{e}$, $y = -\frac{1}{e}$

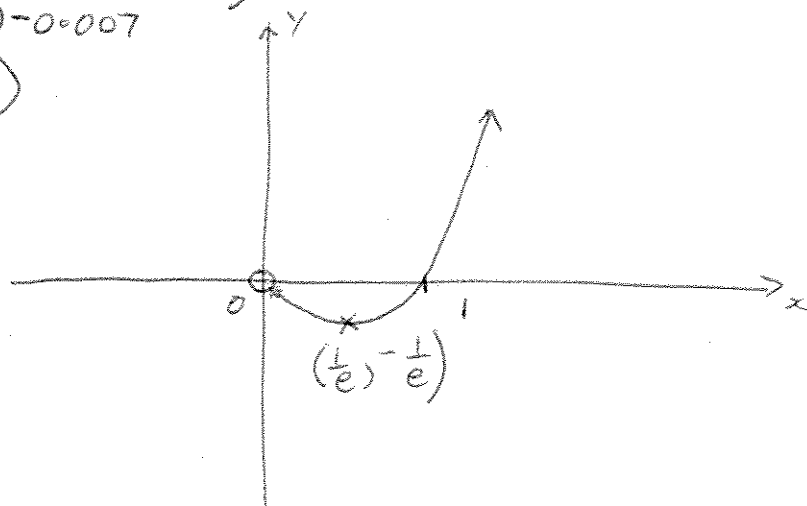
(iii) $y'' = \frac{1}{x}$

> 0 when $x = \frac{1}{e}$

$\therefore (\frac{1}{e}, -\frac{1}{e})$ is a local minimum

(iv) -0.007

(✓)



Q 3

$$(a) \int_{-2}^2 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_{-2}^2$$

$$= \frac{1}{2} (e^4 - e^{-4})$$

$$(b) \frac{d}{dx} (e^{-x^2}) = -2xe^{-x^2}$$

$$\therefore \int_0^1 -2xe^{-x^2} dx = \left[e^{-x^2} \right]_0^1$$

$$\int_0^1 xe^{-x^2} dx = -\frac{1}{2} \left[e^{-x^2} \right]_0^1$$

$$= -\frac{1}{2} (e^{-1} - e^0)$$

$$= -\frac{1}{2} \left(\frac{1}{e} - 1 \right)$$

$$= \frac{1}{2} \left(1 - \frac{1}{e} \right)$$

$$(c) (i) N = N_0 e^{kt}$$

$$\frac{dN}{dt} = k N_0 e^{kt}$$

$$= kN$$

$$(ii) 2N_0 = N_0 e^{200k}, \quad 200k = \log 2$$

$$k = \frac{\log 2}{200}$$

$$(iii) 10N_0 = N_0 e^{\frac{t}{200} \log 2}$$

$$\frac{t}{200} \log 2 = \log 10$$

$$t = \frac{200 \log 10}{\log 2}$$

$$= \frac{21885}{664} \text{ min (nearest minute)}$$

$$\begin{aligned} (d) \int_{-1}^1 x^2 e^x dx &\doteq \frac{1}{3} [y_0 + y_2 + 4y_1] \\ &= \frac{1}{3} (e^{-1} + e + 4e^0) \\ &= \frac{1}{3} (e^{-1} + e + 2) \\ &\approx 1.7 \end{aligned}$$